ITERATIVE ROOT-FINDING METHODS AND THE POLYNOMIAL FACTORIZATION PROCEDURE

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Abstract. This paper implements and analyzes Newton's method, the secant method, and the bisection method to estimate real roots of univariate polynomials. We compare convergence rates, basins of attraction, and failure modes through theoretical and computational analysis. Newton's method achieves quadratic convergence ($q = 2$) for simple roots and linear convergence ($q = 1 - 1/k$) for roots of multiplicity k, whereas the bisection method guarantees linear convergence ($q = 1$) across any sign-changing bracket, and the secant method achieves superlinear convergence of order equal to the golden ratio ($\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$). Finally, by combining Cauchy and Fujiwara root bounds, critical-point partitioning, inflection-guided Newton initialization, and bottom-up dynamic programming across the derivative tower, we introduce the **Polynomial Factorization Procedure (PFP)**—an $\mathcal{O}(n^2)$ algorithm that deterministically isolates and approximates all real roots of any polynomial along with their exact multiplicities.

1. Introduction

A recurring problem in mathematics and scientific computing is to find the roots of polynomials. Computer software systems usually look for roots numerically rather than symbolically, through iterative estimation methods that converge closer and closer to the actual root with each step. We can compare root-finding methods by how fast they converge and by their failure cases; often, choosing the appropriate method involves a trade-off between fast convergence and robustness (having minimal failure cases).

In this paper, we investigate three foundational iterative root-finding methods for polynomials in one variable: Newton's method, the secant method, and the bisection method. For consistency across all three initial examples, we use the polynomial

$$f(x) = \frac{1}{8}x^2 - \frac{1}{2},$$

whose positive root we seek to approximate, denoted $\alpha = 2$.

1.1. Newton's Method

Newton's method accepts an initial guess x_0, evaluates $f(x_0)$ and its derivative $f'(x_0)$, and uses these values to construct the tangent line to the curve at $(x_0, f(x_0))$. The subsequent iterate x_n ($n > 0$) is the x-coordinate where this tangent line crosses the x-axis. We repeat this process until $|f(x_n)|$ is sufficiently close to 0. Figure 1a and Table 1 show the iterations of Newton's method on $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ starting from $x_0 = 4$, terminating when $|e_n| = |x_n - \alpha| < 10^{-5}$.

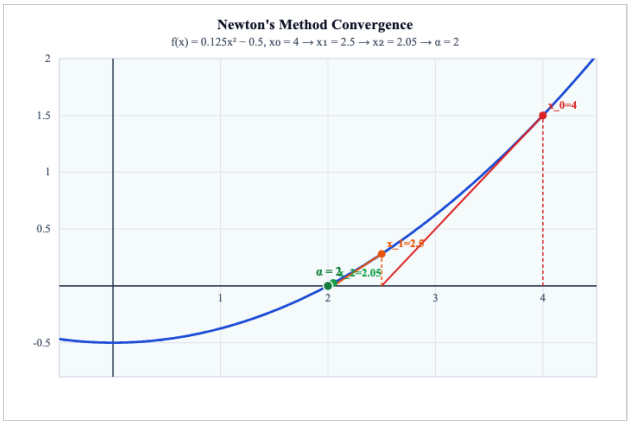


Figure 1a. Successful convergence of Newton's method from $x_0 = 4$ to $\alpha = 2$.

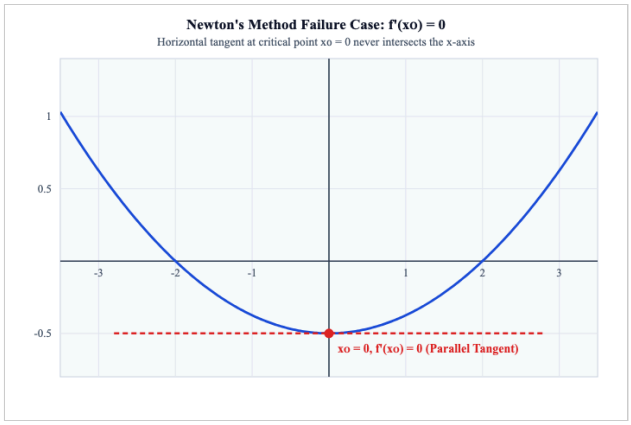


Figure 1b. Newton failure at critical point $x_0 = 0$ where $f'(x_0) = 0$.

Table 1. Newton's Method Iterations for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ ($x_0 = 4$).

Step n	x_n	α	$e_n = x_n - \alpha$	$f(x_n)$
0	4.000000	2	2.000000	1.500000
1	2.500000	2	0.500000	0.281250
2	2.050000	2	0.050000	0.025312
3	2.000610	2	0.000610	0.000305
4	2.000000	2	0.000000	0.000000

We immediately notice that, since Newton's method requires evaluating $f'(x_n)$ at every step in order to draw the tangent line, the method fails if at any point $f'(x_n) = 0$. Visually, $f'(x_n) = 0$ causes the tangent line to run parallel to the x-axis and never intersect it, as illustrated in Figure 1b.

1.2. Secant Method

Unlike Newton's method, which requires analytical derivatives, the Secant method uses two initial guesses, x_0 and x_1, to approximate the tangent line by a secant line. At each step n, the function is evaluated at x_{n-1} and x_n, and the x-intercept of the secant line through $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$ becomes the next approximation x_{n+1}. Figure 2a and Table 2 display the iterations of the Secant method applied to $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ from $(x_0, x_1) = (1, 4)$.

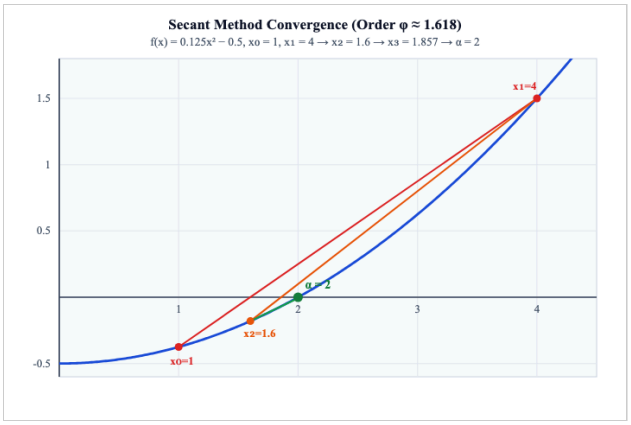


Figure 2a. Secant method convergence from $(x_0, x_1) = (1, 4)$.

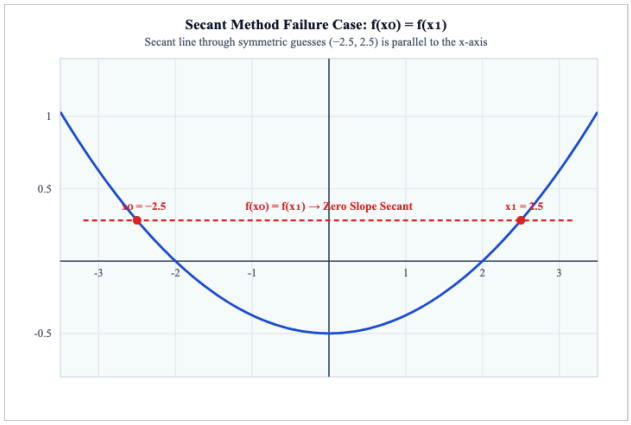


Figure 2b. Secant method failure when $f(x_0) = f(x_1)$.

Table 2. Secant Method Iterations for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ ($x_0 = 1, x_1 = 4$).

Step n	x_n	x_{n+1}	α	$e_n = x_n - \alpha$	$f(x_n)$
0	1.000000	4.000000	2	-1.000000	-0.375000
1	4.000000	1.600000	2	2.000000	1.500000
2	1.600000	1.857143	2	-0.400000	-0.180000
3	1.857143	2.016529	2	-0.142857	-0.068878
4	2.016529	1.999390	2	0.016529	0.008299
5	1.999390	1.999997	2	-0.000610	-0.000305
6	1.999997	2.000000	2	-0.000003	-0.000001

Similarly to Newton's method, the secant method fails when the secant line connecting x_{n-1} and x_n fails to cross the x-axis, which occurs whenever $f(x_{n-1}) = f(x_n)$ (Figure 2b).

1.3. Bisection Method

The Bisection method iteratively narrows down an interval $[l_n, r_n]$ that brackets a root, starting from an initial interval $[l_0, r_0]$ where $f(l_0)$ and $f(r_0)$ have opposite signs ($f(l_0)f(r_0) < 0$). At each step, the midpoint $x_n = \frac{l_n + r_n}{2}$ is evaluated, and comparing the sign of $f(x_n)$ with the signs of $f(l_n)$ and $f(r_n)$ determines which half-interval contains the root. Figure 3 and Table 3 illustrate all 16 iterations of the Bisection method applied to $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ on $[1, 4]$.

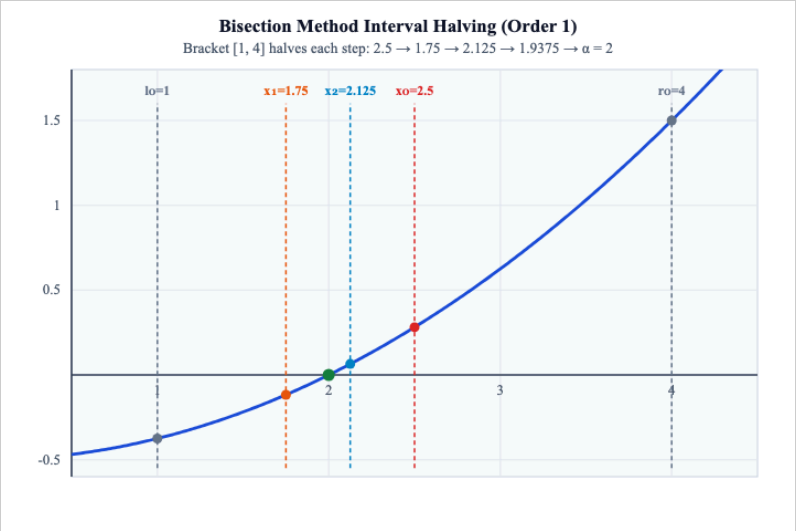


Figure 3. Bisection interval halving on $[1, 4]$ for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$.

Table 3. Bisection Method Iterations for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ on $[1, 4]$.

Step n	l_n	r_n	α	$e_n = x_n - \alpha$	$f(x_n)$
0	1.000000	4.000000	2	0.500000	0.281250
1	1.000000	2.500000	2	-0.250000	-0.117188
2	1.750000	2.500000	2	0.125000	0.064453
3	1.750000	2.125000	2	-0.062500	-0.030762
4	1.937500	2.125000	2	0.031250	0.015747
5	1.937500	2.031250	2	-0.015625	-0.007782
6	1.984375	2.031250	2	0.007812	0.003914
7	1.984375	2.007812	2	-0.003906	-0.001951
8	1.996094	2.007812	2	0.001953	0.000977
9	1.996094	2.001953	2	-0.000977	-0.000488
10	1.999023	2.001953	2	0.000488	0.000244
11	1.999023	2.000488	2	-0.000244	-0.000122
12	1.999756	2.000488	2	0.000122	0.000061
13	1.999756	2.000122	2	-0.000061	-0.000031
14	1.999939	2.000122	2	0.000031	0.000015
15	1.999939	2.000031	2	-0.000015	-0.000008
16	1.999985	2.000031	2	0.000008	0.000004

Even on this single quadratic benchmark $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$, a striking disparity in convergence speed emerges: Newton's method reaches six-decimal accuracy in just 4 iterations (Table 1), the secant method requires 6 iterations (Table 2), and the bisection method requires 16 iterations (Table 3). Conversely, the faster methods exhibit sharper failure modes when initialized poorly—Newton's method can oscillate or diverge and fails whenever $f'(x_n) = 0$,

while the secant method fails whenever $f(x_n) = f(x_{n-1})$. The bisection method exhibits the slowest convergence, but only fails when all roots have even multiplicity, making it the most robust of the three classical algorithms. These trade-offs are summarized in Table 4.

Table 4. Comparative Properties of Iterative Root-Finding Methods.

Method	Order of Convergence	Robustness	Derivative Requirement
Newton	2 (simple roots), 1 (multiple)	Sensitive to x_0, cyclic orbits	Requires $f'(x)$
Secant	$\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$	Fails if $f(x_n) = f(x_{n-1})$	Derivative-free
Bisection	1 (linear halving)	Guaranteed on sign change	Derivative-free
PFP (Ours)	2 **(on isolated brackets)**	**Finds all real roots + multiplicities**	**Uses $f'(x)$ and $f''(x)$**

To rigorously measure and compare the rate at which an iterative sequence approaches a true root α, we formally define the *order of convergence*.

Definition 1.1 (Order of Convergence). *A sequence x_n converging to α has order of convergence $q \geq 1$ if*

$$\lim_{n\to\infty} \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|^q} = \mu$$

for some positive constant $\mu > 0$, called the asymptotic error constant.

Intuitively, an estimate that converges with order $q = 1$ gains a constant number of accurate digits per iteration, whereas an estimate that converges with order $q = 2$ roughly doubles its number of accurate digits at each step. Beyond their local convergence rates, a fundamental weakness shared by all three classical methods is that they solve for only a single local root and provide no global information on how many real roots a polynomial possesses. To address this, we develop a global structural theory of polynomials (Section 2), analyze Newton's rational map Ψ_p (Section 3), prove the convergence orders of the Secant and Bisection methods (Section 4), present the **Polynomial Factorization Procedure (PFP)** (Section 5), and establish tighter root bounds and an $\mathcal{O}(n^2)$ dynamic programming refinement (Section 6).

2. Polynomials, Root Bounds, and Critical Points

In this section, we lay the theoretical foundation for our root-finding algorithms by establishing how the roots and critical points of a polynomial constrain one another. Specifically, if a polynomial $p(x)$ with roots $\alpha_1, \ldots, \alpha_k$ is transformed into $q(x) = c \cdot p(ax + b)$ with $a, c \neq 0$, then $q\left(\frac{\alpha_i - b}{a}\right) = c \cdot p(\alpha_i) = 0$, so every root α_i of p maps to a root $\frac{\alpha_i - b}{a}$ of q. Scaling by $c = 1/a_n$ leaves the roots invariant, allowing us to bound the magnitude of all roots.

Proposition 2.1 (Cauchy ℓ_1 Bound on Root Magnitudes). *Given a polynomial $p(x) = \sum_{r=0}^{n} a_r x^r$ with $a_n \neq 0$, all roots of p lie strictly within $[-M_p, M_p]$, where*

$$M_p = \max\left\{1, \sum_{r=0}^{n-1} \left|\frac{a_r}{a_n}\right|\right\}.$$

Proof. Consider the monic polynomial $q(x) = x^n + \sum_{r=0}^{n-1} b_r x^r$ where $b_r = a_r/a_n$, which has the exact same roots as $p(x)$. For any root α satisfying $|\alpha| \geq 1$, we have $|\alpha|^r \leq |\alpha|^{n-1}$ for all $0 \leq r \leq n-1$. By the triangle inequality, $|\alpha|^n = \left|\sum_{r=0}^{n-1} b_r \alpha^r\right| \leq |\alpha|^{n-1} \sum_{r=0}^{n-1} |b_r|$. Dividing by $|\alpha|^{n-1}$ yields $|\alpha| \leq \sum_{r=0}^{n-1} |a_r/a_n| \leq M_p$. ∎

Proposition 2.1 reduces the search space for all real roots from $(-\infty, \infty)$ to $[-M_p, M_p]$. Next, we examine the critical points of $p(x)$ (roots of $p'(x)$) and show that they are confined within the exact same bound $[-M_p, M_p]$.

Corollary 2.2 (Critical-Point Magnitude Bound). *All critical points of $p(x)$ (roots of $p'(x)$) are also bounded within $[-M_p, M_p]$.*

Proof. Applying Proposition 2.1 to $p'(x) = \sum_{r=1}^{n} r a_r x^{r-1}$ bounds all critical points within $[-M_{p'}, M_{p'}]$, where $M_{p'} = \max\left\{1, \sum_{r=1}^{n-1} \left|\frac{r a_r}{n a_n}\right|\right\} \leq \max\left\{1, \sum_{r=0}^{n-1} \left|\frac{a_r}{a_n}\right|\right\} = M_p$. ∎

Beyond sharing the same outer magnitude bound, roots and critical points interlace and mark strict local boundaries for one another.

Proposition 2.3 (Critical-Point Separation of Real Roots). *There is always at least one critical point strictly between any two distinct real roots of $p(x)$. Moreover, if $p(x)$ has k distinct real roots, it has at least $k-1$ real critical points.*

Proof. Since any polynomial p is continuously differentiable on $\mathbb{R}$, for any two distinct roots $a < b$ with $p(a) = p(b) = 0$, Rolle's Theorem guarantees $c \in (a, b)$ with $p'(c) = \frac{p(b)-p(a)}{b-a} = 0$. Applying this across k sorted roots $\alpha_1 < \cdots < \alpha_k$ yields at least $k-1$ critical points. ∎

While the converse of Proposition 2.3 does not always hold, Proposition 2.3 immediately implies that consecutive critical points partition the real line into monotonic intervals, each containing at most one root.

Corollary 2.4 (Uniqueness of Simple Roots Between Consecutive Critical Points). *There can be at most one real root strictly between any two consecutive critical points (CCPs) $c_1 < c_2$.*

Proof. If two distinct roots $\alpha_1 < \alpha_2$ lay in (c_1, c_2), Proposition 2.3 would guarantee a critical point $c^* \in (\alpha_1, \alpha_2) \subset (c_1, c_2)$, contradicting the assumption that c_1 and c_2 are consecutive critical points. ∎

To determine whether a given pair of CCPs contains a root, we invoke the Intermediate Value Theorem: since $p(x)$ is continuous and strictly monotonic on $[c_1, c_2]$, a simple root $\alpha \in (c_1, c_2)$ exists if and only if $p(c_1)$ and $p(c_2)$ have opposite signs.

Proposition 2.5 (Derivative Sign Invariance on CCP Intervals). *Given a CCP interval (c_1, c_2) containing a simple root α, $p(x)$ and $p'(x)$ have opposite signs on $[c_1, \alpha)$ and identical signs on $(\alpha, c_2]$.*

Proof. By the Intermediate Value Theorem, p changes sign at α while p' has constant nonzero sign on (c_1, c_2). For $x = \alpha + \epsilon \in (\alpha, c_2]$ with $\epsilon > 0$, $p'(\alpha) = \lim_{\epsilon \to 0^+} \frac{p(\alpha+\epsilon)-p(\alpha)}{\epsilon} = \lim_{\epsilon \to 0^+} \frac{p(x)}{\epsilon}$, so $p(x)$ and $p'(x)$ share the same sign on $(\alpha, c_2]$ and opposite signs on $[c_1, \alpha)$. ∎

What if a critical point is itself a root of $p(x)$? At such a point, $p(\alpha) = p'(\alpha) = 0$. We formalize this through root multiplicity and prove that derivative multiplicity coincides with algebraic factor multiplicity.

Definition 2.6 (Root Multiplicity via Higher Derivatives). *A value α is a root of $p(x)$ with multiplicity $k \geq 1$ if $p^{(l)}(\alpha) = 0$ for all $0 \leq l < k$ and $p^{(k)}(\alpha) \neq 0$.*

Lemma 2.7 (Equivalence of Derivative Multiplicity and Factor Multiplicity). *A root α has derivative multiplicity $k \geq 1$ if and only if $p(x) = (x-\alpha)^k q(x)$ with $q(\alpha) \neq 0$.*

Proof. If $p(x) = (x-\alpha)^k q(x)$ with $q(\alpha) \neq 0$, repeated differentiation via the product rule yields $p^{(l)}(\alpha) = 0$ for $0 \leq l < k$ and $p^{(k)}(\alpha) = k!q(\alpha) \neq 0$. Conversely, by induction on k, if α has multiplicity n, then α is a root of $p'(x)$ of multiplicity $n-1$; writing $p(x) = (x-\alpha)^r s(x)$ with $s(\alpha) \neq 0$ gives $p'(x) = (x-\alpha)^{r-1}[rs(x) + (x-\alpha)s'(x)]$, which has multiplicity $r-1 = n-1$, so $r = n$. ∎

3. Newton's Method & Inflection-Guided Basins

Building upon Section 2, we now analyze Newton's recurrence $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ via Newton's rational map $\Psi_p(x)$.

Definition 3.1 (Newton's Rational Map). *Newton's rational function is $\Psi_p(x) = x - \frac{p(x)}{p'(x)}$, with n-fold iterate $\Psi_p^n(x) = \Psi_p(\Psi_p^{n-1}(x))$.*

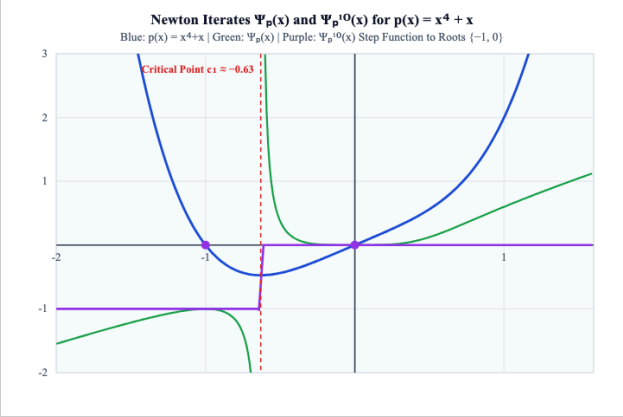


Figure 4. $p(x) = x^4 + x$ (blue), $\Psi_p(x)$ (green), and $\Psi_p^{10}(x)$ (purple).

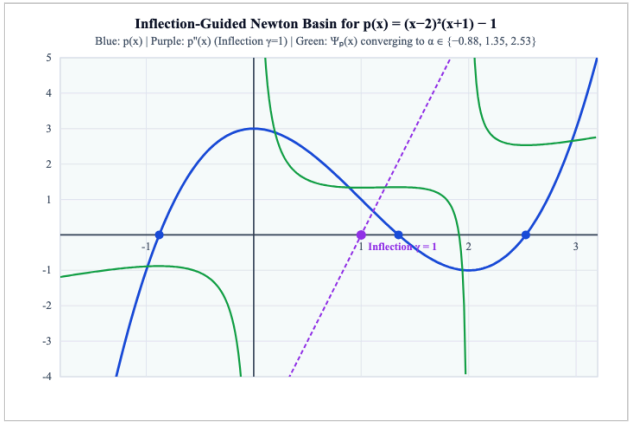


Figure 5. $p(x) = (x-2)^2(x+1) - 1$, $p''(x)$ (purple), and $\Psi_p(x)$ (green).

Proposition 3.2 (Fixed-Point Consistency of Newton's Sequence). *If $x_n = \Psi_p^n(x_0) \to \alpha \in \mathbb{R}$, then α is a root of $p(x)$.*

Proof. Rearranging $x_{n+1} = x_n - p(x_n)/p'(x_n)$ yields $p(x_n) = p'(x_n)(x_n - x_{n+1})$. As $n \to \infty$, $x_n - x_{n+1} \to 0$ and $p'(x_n) \to p'(\alpha)$ by continuity, so $p(\alpha) = \lim_{n\to\infty} p(x_n) = 0$. ∎

Lemma 3.3 (Monotone Basin Convergence Lemma). *If p is monotonic and Ψ_p is monotonically increasing on $[a, b]$ containing a simple root α, then $\forall x_0 \in (a, b)$, $\Psi_p^n(x_0) \to \alpha$ monotonically.*

Proof. For $x \in (a, \alpha)$, Proposition 2.5 gives $p(x)/p'(x) < 0$, so $x < \Psi_p(x) \leq \Psi_p(\alpha) = \alpha$. Thus $\Psi_p^n(x)$ is monotonically increasing and bounded above by α. Symmetrically, for $x \in (\alpha, b)$, $p(x)/p'(x) > 0 \implies \alpha \leq \Psi_p(x) < x$, so $\Psi_p^n(x)$ is monotonically decreasing and bounded below by α. By the Monotone Convergence Theorem and Proposition 3.2, both converge to α. ∎

Proposition 3.4 (Guaranteed Convergent Sub-Interval). *Partitioning any sign-changing CCP bracket $[c_\alpha, c'_\alpha]$ by inflection points $p''(\gamma_i) = 0$ isolates a sub-interval $[\gamma_t, \gamma_{t+1}]$ containing α where $\operatorname{sign}(p(x_0)) = \operatorname{sign}(p''(x_0))$ at one endpoint x_0, guaranteeing $\Psi'_p(x) = \frac{p(x)p''(x)}{(p'(x))^2} > 0$ and monotonic convergence $\Psi_p^n(x_0) \to \alpha$.*

Proof. Let $s = \operatorname{sign}(p'(x)) \in \{+1, -1\}$ on $I_\alpha = [c_\alpha, c'_\alpha]$. On the unique sign-changing inflection sub-interval $[\gamma_t, \gamma_{t+1}]$ containing α, p has sign $-s$ on $[\gamma_t, \alpha)$ and $+s$ on $(\alpha, \gamma_{t+1}]$, while p'' has constant nonzero sign. If $\operatorname{sign}(p'') = +s$, then p and p'' share the same sign on $(\alpha, \gamma_{t+1}]$; if $\operatorname{sign}(p'') = -s$, they share the same sign on $[\gamma_t, \alpha)$. In either half-interval, $\Psi'_p(x) = \frac{p(x)p''(x)}{(p'(x))^2} > 0$, so Lemma 3.3 guarantees monotonic convergence to α. ∎

Algorithm 1: Inflection-Guided Initial Guess for Newton's Method

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1. **Input:** Polynomial $p(x)$, consecutive critical points $c_1 < c_2$ with $p(c_1)p(c_2) < 0$, sorted inflection points $D = (\gamma_1, \ldots, \gamma_r) \subset (c_1, c_2)$.
2. Let $s = \operatorname{sign}(p'((c_1 + c_2)/2))$ and form partition $P = [c_1, \gamma_1, \ldots, \gamma_r, c_2]$.
3. For each adjacent pair (a, b) in P with $p(a)p(b) \le 0$: if $\operatorname{sign}(p''((a+b)/2)) = s$, return $x_0 = b$; else return $x_0 = a$.

Theorem 3.5 (Order of Convergence of Newton's Method). *When $x_n = \Psi_p^n(x_0) \to \alpha$ for a root α of multiplicity $k \ge 1$, Newton's method converges quadratically ($q = 2$) with $\mu = \left|\frac{p''(\alpha)}{2p'(\alpha)}\right|$ when $k = 1$, and linearly ($q = 1$) with rate $1 - 1/k$ when $k > 1$.*

Proof. Let $e_n = x_n - \alpha$. Expanding $p(x_n)$ and $p'(x_n)$ via Taylor series around a root α of multiplicity k (so $p^{(l)}(\alpha) = 0$ for $0 \le l < k$ and $p^{(k)}(\alpha) \ne 0$) yields

$$e_{n+1} = e_n - \frac{\frac{p^{(k)}(\alpha)}{k!}e_n^k + \frac{p^{(k+1)}(\alpha)}{(k+1)!}e_n^{k+1} + \mathcal{O}(e_n^{k+2})}{\frac{p^{(k)}(\alpha)}{(k-1)!}e_n^{k-1} + \frac{p^{(k+1)}(\alpha)}{k!}e_n^k + \mathcal{O}(e_n^{k+1})} = e_n\left(1 - \frac{1}{k}\right) + e_n^2 \frac{p^{(k+1)}(\alpha)}{k(k+1)p^{(k)}(\alpha)} + \mathcal{O}(e_n^3).$$

When $k = 1$, the linear term $e_n(1 - 1/k)$ vanishes and $\lim_{n\to\infty} \frac{|e_{n+1}|}{|e_n|^2} = \left|\frac{p''(\alpha)}{2p'(\alpha)}\right|$ (order $q = 2$). When $k > 1$, $\lim_{n\to\infty} \frac{|e_{n+1}|}{|e_n|} = 1 - \frac{1}{k}$ (order $q = 1$). ∎

4. Secant Method & Bisection Method

Theorem 4.1 (Golden-Ratio Convergence of the Secant Method). *If the secant sequence $x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$ converges to a simple root α with $f'(\alpha), f''(\alpha) \ne 0$, its order of convergence is the golden ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618034$.*

Proof. Let $e_n = x_n - \alpha$. Substituting $x_n = \alpha + e_n$ into the secant recurrence and expanding $f(\alpha + e_n) = f'(\alpha)e_n + \frac{1}{2}f''(\alpha)e_n^2 + \mathcal{O}(e_n^3)$ yields $e_{n+1} = \left(\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}\right)\left(\frac{f(x_n)/e_n - f(x_{n-1})/e_{n-1}}{e_n - e_{n-1}}\right) e_n e_{n-1} \approx Ce_n e_{n-1}$ where $C = \frac{f''(\alpha)}{2f'(\alpha)}$. Setting $g_n = \log|Ce_n|$ yields the Fibonacci recurrence $g_{n+1} \approx g_n + g_{n-1}$, so $\lim_{n\to\infty} \frac{g_{n+1}}{g_n} = \phi = \frac{1+\sqrt{5}}{2}$ and $\lim_{n\to\infty} \frac{|e_{n+1}|}{|e_n|^\phi} = |C|^{\phi-1}$. ∎

Theorem 4.2 (Linear Convergence of Bisection). *On any bracket* $[l_0, r_0]$ *with* $f(l_0)f(r_0) < 0$*, bisection satisfies* $r_n - l_n = (r_0 - l_0)/2^n$ *and converges with order* $q = 1$ *and rate* $\mu = 1/2$*.*

Proof. Starting from $[l_0, r_0]$, the bisection method halves the bracket width at every iteration, so $r_n - l_n = \frac{r_0 - l_0}{2^n}$. Thus the interval error ratio satisfies $\lim_{n\to\infty} \frac{r_{n+1}-l_{n+1}}{r_n - l_n} = \frac{(r_0-l_0)/2^{n+1}}{(r_0-l_0)/2^n} = \frac{1}{2}$, establishing linear convergence of order $q = 1$ in $\mathcal{O}(\log_2((r_0 - l_0)/\varepsilon))$ steps. ∎

5. Polynomial Factorization Procedure (PFP)

Algorithm 2: Polynomial Factorization Procedure (PFP)

1. **Base Case:** If $\deg(f) \in \{1, 2\}$, solve for real roots and multiplicities directly via closed-form linear/quadratic formulas.
2. **Recursive Derivative Calls:** Compute critical points $\{(c_i, m_i)\} = \mathrm{PFP}(f')$ and inflection points $\{d_j\} = \mathrm{PFP}(f'')$.
3. **Multiple Root Detection:** For each (c_i, m_i), if $|f(c_i)| = 0$, record c_i as a root of f with multiplicity $m_i + 1$.
4. **Simple Root Isolation:** Partition $[-M_f, M_f]$ into monotone intervals $(-M_f, c_1), (c_1, c_2), \ldots, (c_k, M_f)$. On each sign-changing interval, select x_0 via Algorithm 1 using inflection points D_j and run safeguarded Newton-Bisection to converge to the unique simple root α (multiplicity 1).

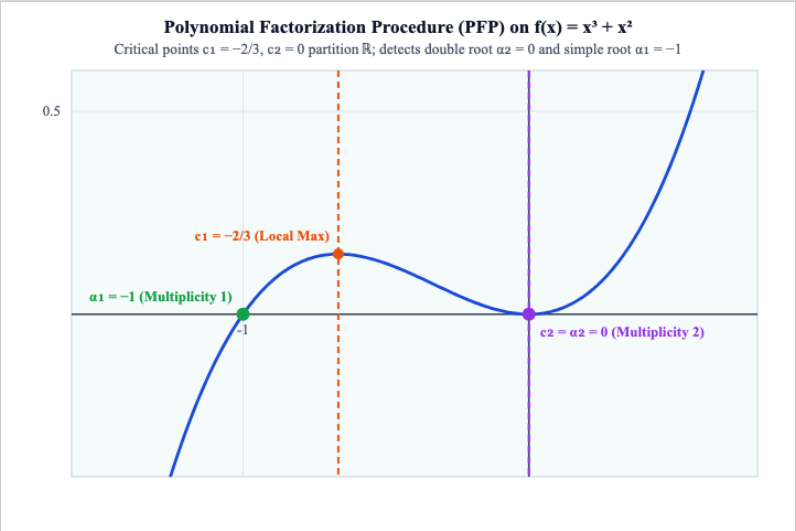


Figure 6. PFP partitioning $f(x) = x^3 + x^2$ by critical points $c_1 = -2/3$ and $c_2 = 0$, detecting double root $\alpha_2 = 0$ ($m = 2$) and simple root $\alpha_1 = -1$ ($m = 1$).

Theorem 5.1 (Completeness of PFP). *For any real polynomial* $f(x)$*, the Polynomial Factorization Procedure isolates and approximates all real roots along with their exact multiplicities.*

Proof. By Proposition 2.1, all real roots lie in $(-M_f, M_f)$. By Lemma 2.7, every multiple root ($m > 1$) is a root of $f'(x)$ of multiplicity $m - 1$ and is therefore identified at the critical-point check $|f(c_i)| = 0$. By Corollary 2.4, at most one simple root lies strictly between consecutive critical points (c_i, c_{i+1}), and since f is strictly monotonic on (c_i, c_{i+1}), a simple root exists in (c_i, c_{i+1}) iff f changes sign across the interval. Proposition 3.4 then guarantees

that Algorithm 1 selects an initial guess in the monotone basin of attraction, ensuring quadratic convergence to each simple root. ■

6. Algorithmic Optimizations, Tighter Bounds, and Complexity Analysis

Proposition 6.1 (ℓ_∞ Cauchy and Fujiwara Radical Root Bounds). *For any polynomial $p(x) = \sum_{r=0}^{n} a_r x^r$ ($a_n \neq 0$), every root $\alpha \in \mathbb{C}$ satisfies $|\alpha| \leq \min\{M_p, M_\infty(p), M_{\mathrm{F}}(p)\}$, where*

$$M_\infty(p) = 1 + \max_{0 \leq r \leq n-1} \left| \frac{a_r}{a_n} \right|, \qquad M_{\mathrm{F}}(p) = 2 \max \left\{ \left| \frac{a_{n-1}}{a_n} \right|, \left| \frac{a_{n-2}}{a_n} \right|^{\frac{1}{2}}, \ldots, \left| \frac{a_1}{a_n} \right|^{\frac{1}{n-1}}, \left| \frac{a_0}{2a_n} \right|^{\frac{1}{n}} \right\}.$$

Proof. Let $b_r = |a_r/a_n|$ and $B = \max_{0 \leq r < n} b_r$. For $|\alpha| > 1$, $|\alpha|^n \leq B \sum_{r=0}^{n-1} |\alpha|^r = B \frac{|\alpha|^n - 1}{|\alpha| - 1} < B \frac{|\alpha|^n}{|\alpha| - 1} \implies |\alpha| < 1 + B = M_\infty(p)$. For Fujiwara's bound $M_{\mathrm{F}}(p)$, if $|\alpha| > M_{\mathrm{F}}(p)$, then $b_{n-k} < (|\alpha|/2)^k$ for $1 \leq k \leq n-1$ and $b_0 < 2(|\alpha|/2)^n$, giving $|\alpha|^n \leq \sum_{k=1}^{n} b_{n-k} |\alpha|^{n-k} < |\alpha|^n (\sum_{k=1}^{n-1} 2^{-k} + 2 \cdot 2^{-n}) = |\alpha|^n$, a contradiction. ■

Proposition 6.2 (Inflection-Point Tangent Landing). *Let (c_1, c_2) be a CCP interval containing a simple root α and a unique inflection point $\gamma \in (c_1, c_2)$ with $p(\gamma) \neq 0$. Then $|p'(x)|$ attains its strict maximum on $[c_1, c_2]$ at $x = \gamma$, and a single Newton step $x_1 = \Psi_p(\gamma) = \gamma - p(\gamma)/p'(\gamma)$ lands strictly inside the Fourier monotone basin where $\operatorname{sign}(p(x_1)) = \operatorname{sign}(p''(x_1))$, guaranteeing $\Psi_p^n(\gamma) \to \alpha$.*

Proof. Since $p''(\gamma) = 0$ and p' has constant sign $s \in \{+1, -1\}$ on (c_1, c_2) while vanishing at c_1, c_2, $|p'(\gamma)| > 0$ is maximized at γ. By Taylor's Theorem with Lagrange remainder around γ, $0 = p(\alpha) = p(\gamma) + p'(\gamma)(\alpha - \gamma) + \frac{1}{2} p''(\xi)(\alpha - \gamma)^2$ for some ξ between γ and α. Dividing by $p'(\gamma)$ yields $x_1 - \alpha = \frac{p''(\xi)}{2p'(\gamma)} (\alpha - \gamma)^2$, so x_1 lies on the side of α where $\operatorname{sign}(p(x_1)) = s \cdot \operatorname{sign}(x_1 - \alpha) = \operatorname{sign}(p''(\xi)) = \operatorname{sign}(p''(x_1))$, where $\Psi_p' > 0$. ■

Algorithm 3: Bottom-Up Dynamic Programming PFP ($\mathcal{O}(n^2)$ Root Isolation)

1. **Derivative Tower:** Precompute $P_k(x) = f^{(k)}(x)$ for $k = 0, 1, \ldots, n-1$, solve the linear base case $P_{n-1}(x) = 0$ to obtain R_{n-1}, and set $R_n = \emptyset$.
2. **Bottom-Up Pass ($k = n-2$ down to 0):** Using critical points $C = R_{k+1}$ and inflection points $D = R_{k+2}$ already computed at the previous two levels, compute the tight bound $M = \min\{M_{P_k}, M_\infty(P_k), M_{\mathrm{F}}(P_k)\} + 1$, record multiple roots $(c_i, m_i + 1)$ where $|P_k(c_i)| < \varepsilon$, and isolate simple roots in each sign-changing interval $(c_i, c_{i+1}) \subset [-M, M]$ via D and safeguarded Newton-Bisection to form R_k.
3. **Output:** Return R_0.

Theorem 6.3 (Quadratic $\mathcal{O}(n^2)$ Complexity of Bottom-Up PFP). *Algorithm 3 reduces the number of derivative levels solved from $\Theta(\phi^n)$ to $n-1$, performing at most $\sum_{k=0}^{n-2} (n-k) = \frac{n(n+1)}{2} - 1 = \mathcal{O}(n^2)$ root-isolation bracket solves.*

Proof. In naive recursive PFP, level k invokes both $\mathrm{PFP}(P_{k+1})$ and $\mathrm{PFP}(P_{k+2})$, satisfying the Fibonacci recurrence $T(n) = T(n-1) + T(n-2) = \Theta(\phi^n)$. Because $P_{k+2} = (P_{k+1})'$, storing and reusing R_{k+1}

and R_{k+2} ensures each derivative P_k of degree $d = n - k$ is partitioned into at most d monotone intervals exactly once, summing to $\sum_{d=2}^{n} d = \frac{n(n+1)}{2} - 1 = \mathcal{O}(n^2)$. ■

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